

# Miguel Zumalacárregui

Group Leader, Astrophysical and Cosmological Relativity — Max Planck Institute for  
Gravitational Physics (Albert Einstein Institute), Potsdam

Ramón y Cajal Fellow — Instituto de Física Teórica IFT-UAM/CSIC, Madrid

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The publication record is maintained on [INSPIRE-HEP](#); research [projects and funding](#) are listed on the website.

## Positions

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- **Ramón y Cajal Fellow**

Instituto de Física Teórica IFT-UAM/CSIC, Madrid · 2026–

- **Group Leader**, Astrophysical and Cosmological Relativity

Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam · 2020–

- **Marie Skłodowska-Curie Global Fellow**

Berkeley Center for Cosmological Physics, with return phase at IPhT Saclay · 2017–2020

- **Nordita Fellow**

Nordic Institute for Theoretical Physics, Stockholm · 2015–2017

- **Postdoctoral Researcher**

Institut für Theoretische Physik, University of Heidelberg · 2013–2015

- **Postdoctoral Researcher**

Instituto de Física Teórica IFT-UAM/CSIC, Madrid · 2013

## Education

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- **PhD in Physics**, *Cum Laude* with International Mention

Universidad Autónoma de Madrid · 2012 · Graduate work at ICC Barcelona and IFT-UAM/CSIC (FPI fellowship)

Dissertation: [Probing the Foundations of the Standard Cosmological Model](#) · [defense presentation](#)

- **Master's thesis**

Universidad Autónoma de Madrid · [PDF](#)

## Supervision

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### Postdoctoral researchers

- **Daniel Ballard** — from October 2026

- **Han-Gil Choi** — 2026–

- **Xikai Shan** — Affiliated postdoctoral researcher, 2026–

- **Juan Urrutia** — from October 2026

- **Héctor Villarrubia-Rojo** — from October 2026

- **Srashti Goyal** — 2023–2026 → Independent Fellow at IUCAA

- **Serena Giardino** — 2024–2025 → Postdoc at Imperial College London

- **Guilherme Brando** — Humboldt Fellow, 2022–2024 → Tenured Researcher at UFES, Brazil

- **Giovanni Tambalo** — 2020–2023 → Postdoc at ETH Zürich, then Marie Skłodowska-Curie Fellow at IPhT Saclay

### PhD students

- **Emanuela Alberici** — from October 2026

- **Manon van Zyl** — from October 2026
- **Nicola Menadeo** — PhD student, U. Naples (co-supervised), 2023–2025, long-term stay → Postdoc at Humboldt University of Berlin
- **Stefano Savastano** — PhD student (co-supervised with A. Buonanno), 2020–2024 → PhD awarded at HU Berlin

### Master’s students

**Matteo Pagani** (U. Bologna, co-supervised, 2026); **Louis Salimbeni** (ETH Zürich, co-supervised, 2026); **Pablo Vega del Castillo** (LMU, 2026); **Shashwat Singh** (Paris Observatory, 2022–2023 → PhD at the University of Glasgow); **Julius Streibert** (FU Berlin, co-supervised, 2022–2023); **Richard Stiskalek** (LMU, co-supervised, 2021–2022 → PhD at Oxford, pre-doctoral Fellow at CCA, now postdoc at Oxford); **Alessandro Longo** (U. Trieste, co-supervised, 2020–2021 → PhD student at SISSA).

### Research interns and undergraduates

**Selin Üstündag** (2026 → PhD at Newcastle University); **Marcel Glätzer** (HU Berlin, 2024–2025); **Jaime Raynaud** (TU Berlin & KTH, 2024–2025 → PhD at MPIA); **Vaibhav Omble** (LMU, 2023–2024); **Daryna Yushchenko** (Princeton summer program, 2024); **Lyla Choi** (Princeton summer program, 2023 → PhD at Stanford).

Visiting researchers, and the students mentored informally before the group started in 2020, are listed at [miguelzuma.github.io/group.html](https://miguelzuma.github.io/group.html).

### Scientific coordination

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- **LISA Consortium**  
member since 2018  
Council member and Senior Co-Chair of the Cosmology Working Group (2025–); section coordinator for gravitational-wave lensing in the LISA Cosmology white paper (2022)
- **Euclid Consortium**  
member since 2014  
Lead of the Science Working Group on Gravitational Waves and of its “bright sirens” work package (2025–); member of the Strong Lensing, Cosmology Theory, and Supernovae & Transients working groups
- **GW-Space 2050**  
Coordinator for the cosmography section, defining ESA’s gravitational-wave missions after LISA (2023–)
- **Multi-messenger lensing white paper**  
Coordinator for the microlensing section (2024)

### Workshops and meetings organized

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- [13th LISA Cosmology Working Group Workshop](#)  
Barcelona, 2026
- [12th LISA Cosmology Working Group Workshop](#)  
Tallinn, 2025
- [Lensing and Wave Optics in Strong Gravity](#)  
Erwin Schrödinger Institute, Vienna, 2024
- [LISA joint Fundamental Physics + Waveform meeting](#)  
Potsdam, 2024
- [ACR Division Seminars](#)  
Max Planck Institute for Gravitational Physics, 2020–2022
- BCCP Cosmology Lunch

UC Berkeley, 2016–2017

- Cosmology Seminars, Institut für Theoretische Physik  
University of Heidelberg, 2014–2015
- TRR33 Winter Schools  
Passo del Tonale, 2013, 2014 and 2015

## Teaching

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### Graduate schools

- **Gravitational waves**, TAE (Taller de Altas Energías), Benasque, 2024  
[Slides](#)
- **Tests of gravity and dark energy with cosmology and gravitational waves**  
II Mesoamerican Workshop on Gravity and Cosmology, Chiapas, 2018
- **Alternative gravity theories**, Cosmology School in the Canary Islands, 2017  
[Video lecture](#)
- **School for Cosmology Tools**, IFT Madrid, 2017  
Lectures on CLASS, hi\_class and MontePython; material on the [Talks & Teaching page](#)
- **Workshop on CLASS and MontePython**, MPIA Munich, 2014

### University courses

- **Berkeley Connect Instructor**, UC Berkeley, 2018–2019  
Two semesters of undergraduate mentoring and teaching
- **Lecturer**, University of Heidelberg, 2014–2015  
Advanced Cosmology; Introduction to CMB Physics; Statistical Mechanics; advanced seminar
- **Teaching assistant**, University of Heidelberg, 2014–2015  
Theoretical Astrophysics (graduate), with M. Bartelmann
- **Teaching assistant**, Universidad Autónoma de Madrid, 2013  
Gravitation and Cosmology; Physics of the Cosmos
- **Mini-course on inhomogeneous cosmologies**, University of Barcelona, 2012

## Awards and fellowships

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- ERC Consolidator Grant (2025): GLOW, Gravitational Lensing of Waves.
- Ramón y Cajal Fellowship (2024 call, awarded 2025).
- Max Planck Society Annual Project (2024): With Einstein on Crooked Paths.
- R3 certification, Spanish Research Agency.
- Berkeley Connect Fellow, UC Berkeley (2018).
- Marie Skłodowska-Curie Global Fellowship (2015).
- Nordita Fellowship (2015).
- Yggdrasil Fellowship (2011): research stay at ITA Oslo.
- FPI doctoral fellowship (2008) and UAM Master's scholarship (2007).

## Evaluation and peer review

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### Evaluation panels

- NASA Hubble Fellowship Program, Physics and Cosmology panel.
- FCT (Portuguese Foundation for Science and Technology), Individual Support Call.
- Max Planck Institute for Gravitational Physics, postdoctoral hiring committee, since 2020.

## External evaluator for funding agencies

European Research Council (Starting Grants), STFC/UKRI, Swiss National Science Foundation, Alexander von Humboldt Foundation, InterTalentum (Marie Skłodowska-Curie COFUND), Netherlands Research Council (NWO), Research Grants Council of Hong Kong, and the Estonian Research Council.

## Refereeing

Peer review for more than 40 papers, including Physical Review Letters, Physical Review D, Nature Astronomy, JCAP, The Astrophysical Journal and Astrophysical Journal Letters.

## Selected talks

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- **IAS and Princeton University**, gravitational-wave meeting, Princeton, 2026 · seminar  
GW231123 as a diffracted and magnified black-hole merger
- **Spanish & Portuguese Relativity Meeting**, Universidad de Murcia, 2026 · plenary  
Gravitational lensing of waves: a new window into astrophysics, dark matter and gravity
- **Leibniz Institute for Astrophysics**, Potsdam, 2024 · institute colloquium  
Gravitational lensing of waves
- **New Frontiers in Strong Gravity**, Benasque, 2024 · plenary  
Wave-optical phenomena in gravitational-wave lensing
- **Workshop on Tensions in Cosmology**, Corfu, 2022 · plenary  
Towards solutions to the Hubble problem beyond Einstein's gravity
- **EuCAPT Annual Symposium**, CERN, 2022 · plenary review  
[Dark energy and tests of gravity with cosmology and gravitational-wave propagation](#)
- **Institute of Theoretical Astrophysics**, University of Oslo, 2021 · institute colloquium  
The dark Universe in the era of gravitational-wave astronomy
- **Progress in Old and New Themes in Cosmology**, Palais des Papes, Avignon, 2020 · plenary  
Tests of gravity with gravitational waves
- **COSMO**, Aachen University, 2019 · plenary  
Tests of dark energy and modified gravity
- **Rencontres du Vietnam**, Very High Energy Phenomena in the Universe, 2018 · plenary  
The dark Universe in the era of gravitational-wave astronomy

## Languages

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Spanish (native), English (C2), German (C1).

## Selected publications

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Chosen for what they opened up rather than for where they appeared. The complete record is on [INSPIRE-HEP](#).

- [Across the Universe: GW231123 as a magnified and diffracted black hole merger](#)  
S. Goyal, H. Villarrubia-Rojo, M. Zumalacárregui · Astrophys. J. Lett. 1008, L12 (2026)
- [Black holes as telescopes: discovering supermassive binaries through quasi-periodic lensed starlight](#)  
H. Wang, M. Zumalacárregui, B. Kocsis · Phys. Rev. Lett. 136, 061403 (2026) · Editors' Suggestion
- [GLoW: novel methods for wave-optics phenomena in gravitational lensing](#)  
H. Villarrubia-Rojo, S. Savastano, M. Zumalacárregui, L. Choi, S. Goyal, L. Dai, G. Tambalo · Phys. Rev. D 111, 103539 (2025)
- [Gravitational wave lensing as a probe of halo properties and dark matter](#)  
G. Tambalo, M. Zumalacárregui, L. Dai, M. Cheung · Phys. Rev. D 108, 103529 (2023)

- [Gravitational wave lensing beyond general relativity: birefringence, echoes and shadows](#)  
J. M. Ezquiaga, M. Zumalacárregui · Phys. Rev. D 102, 124048 (2020) · Editors' Suggestion
- [Gravity in the era of equality: towards solutions to the Hubble problem without fine-tuned initial conditions](#)  
M. Zumalacárregui · Phys. Rev. D 102, 023523 (2020)
- [Limits on stellar-mass compact objects as dark matter from gravitational lensing of type Ia supernovae](#)  
M. Zumalacárregui, U. Seljak · Phys. Rev. Lett. 121, 141101 (2018) · Editors' Suggestion
- [Dark energy after GW170817: dead ends and the road ahead](#)  
J. M. Ezquiaga, M. Zumalacárregui · Phys. Rev. Lett. 119, 251304 (2017) · Editors' Suggestion
- [hi\\_class: Horndeski in the Cosmic Linear Anisotropy Solving System](#)  
M. Zumalacárregui, E. Bellini, I. Sawicki, J. Lesgourgues, P. G. Ferreira · JCAP 08, 019 (2017)
- [Transforming gravity: from derivative couplings to matter to second-order scalar–tensor theories beyond the Horndeski Lagrangian](#)  
M. Zumalacárregui, J. García-Bellido · Phys. Rev. D 89, 064046 (2014)